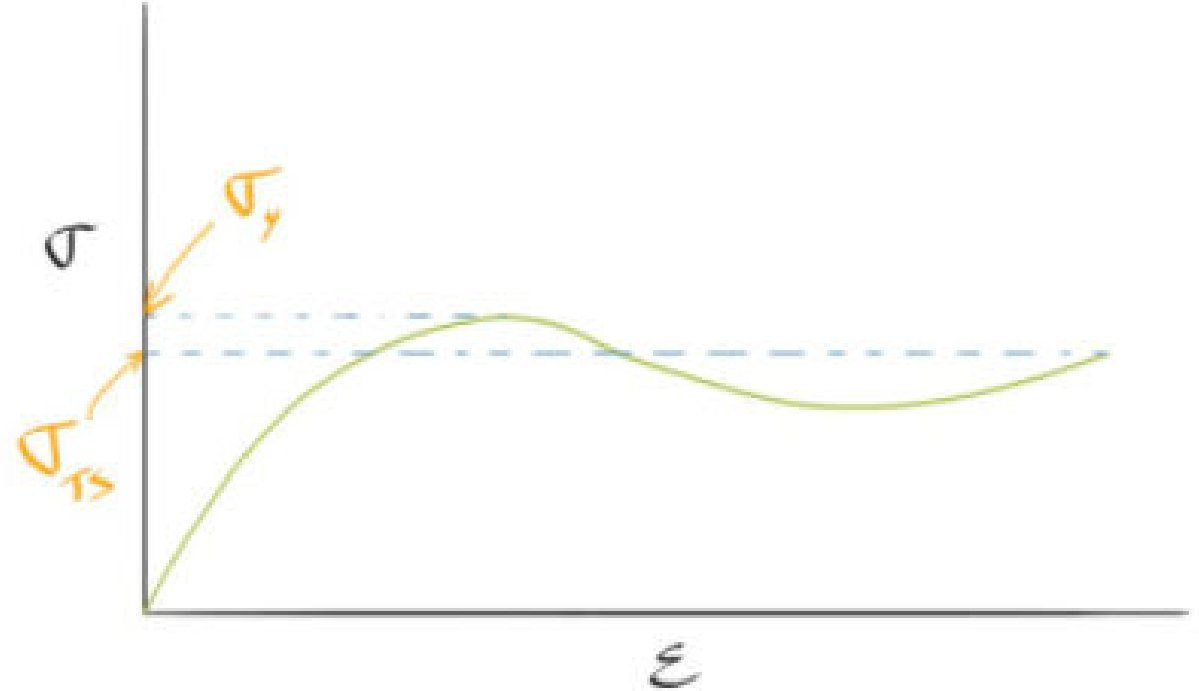
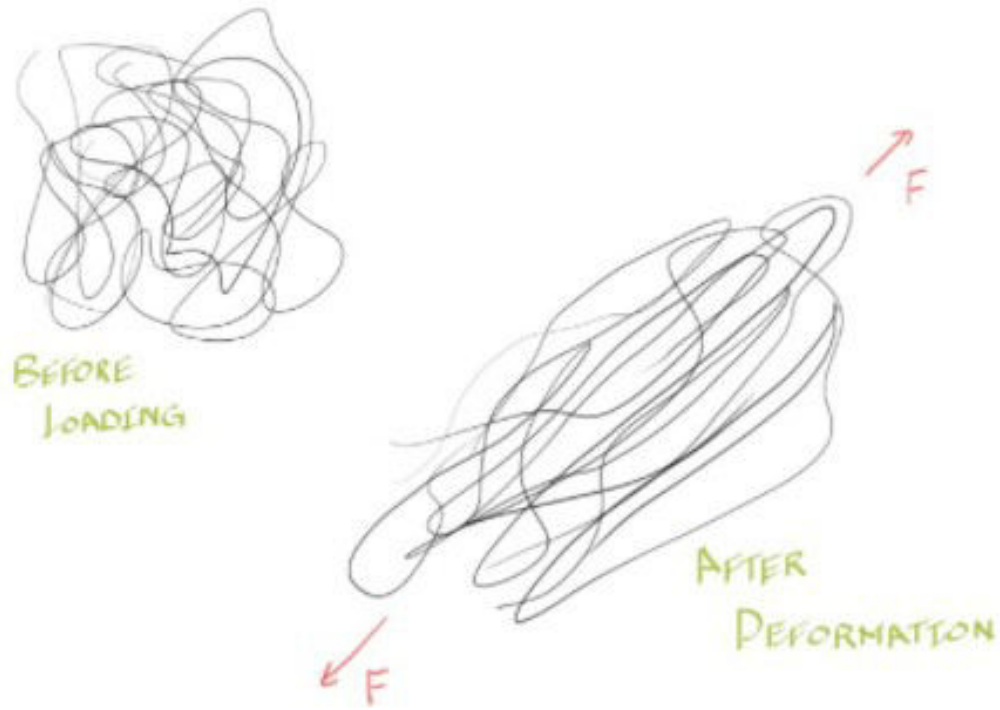


APS110 Tutorial #8

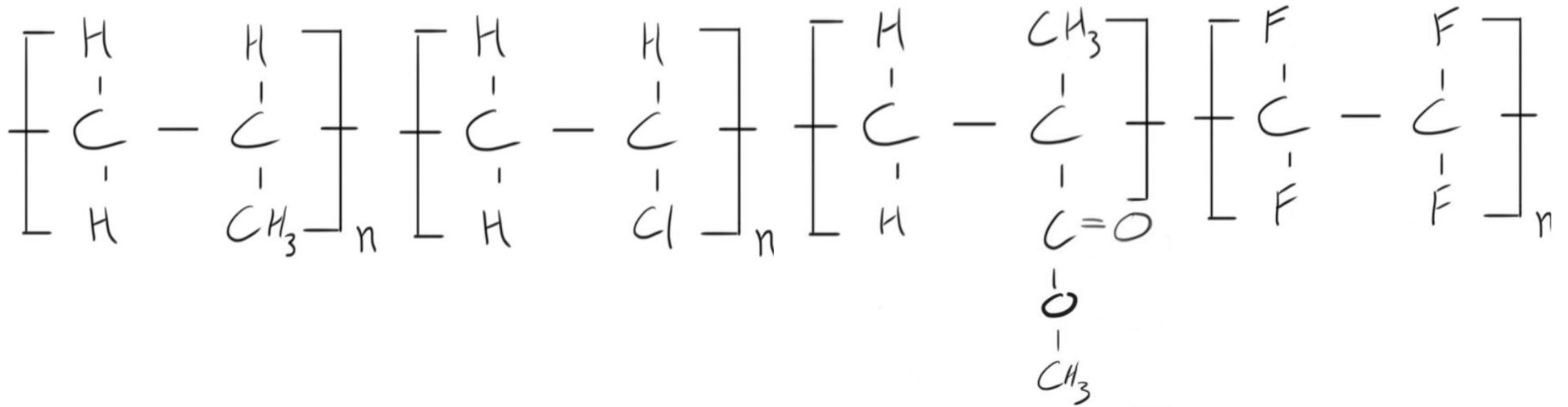
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Review – Polymer Materials



Review – Mer Units



Review – Molecular Averages

Number Average Mol. Weight

$$\bar{M}_n = \sum_{n=1}^i M_n x_n$$

Gives you a sense of the average
molecular length of the polymer
Gives you a sense of the average
molecular length of the polymer

Weight Average Mol. Weight

$$\bar{M}_w = \sum_{n=1}^i M_n w_n$$

Gives you a sense of the average
molecular weight of the polymer
Gives you a sense of the average
molecular weight of the polymer

Indication of how broad the distribution is
Indication of how broad the distribution is

$$D = \frac{\bar{M}_w}{\bar{M}_n}$$

Practice Question 1

1. A hypothetical polymer sample consists of the following molecules
 - a) 15 molecules having an average molecular weight of 11,000 g/mol
 - b) 23 molecules having an average molecular weight of 23,000 g/mol
 - c) 38 molecules having an average molecular weight of 42,000 g/mol
 - d) 29 molecules having an average molecular weight of 64,000 g/mol

Calculate both the number average and weight average molecular weight

Practice Question 1

$$\bar{M}_n' = \frac{\sum_{n=1}^i M_n \sum_{n=1}^i M_n x_n}{\sum_{n=1}^i M_n} = \frac{15 \cdot 11e3 + 23 \cdot 23e3 + 38 \cdot 42e3 + 29 \cdot 64e3}{15 + 23 + 38 + 29} = 39.5e3 \text{ g/mol}$$

$$\bar{M}_w' = \frac{\sum_{n=1}^i M_n \sum_{n=1}^i M_n^2 x_n}{\sum_{n=1}^i M_n^2} = \frac{(15 \cdot 11e3) \cdot 11e3 + (23 \cdot 23e3) \cdot 23e3 + (38 \cdot 42e3) \cdot 42e3 + (29 \cdot 64e3) \cdot 64e3}{15^2 \cdot 11e3 + 23^2 \cdot 23e3 + 38^2 \cdot 42e3 + 29^2 \cdot 64e3} = 48e3 \text{ g/mol}$$

Practice Question 2

2. Explain all the processes or microstructural changes that may be used to increase the strength of a polymer. Consider bonding, chain length, crystallinity, etc.
- Increase chain length (molecular weight)
 - More entanglement
 - Introduce polar bonds (dipoles) into mer units
 - Stronger 2° bonds
 - Increase crystallinity
 - Fewer, short branches allow molecules to pack closer
 - Crystalline regions are held together better by 2° bonds
 - Increase cross-linking
 - Strong 1° bonds between chains
 - Prevent/minimize chain sliding
 - Very cross-linked polymers become network polymers

Practice Quiz

1. Which of the following is most likely to be correct when comparing the density for samples of polyethylene?
 - a) Highly crystalline $\approx$ low crystallinity, high molecular weight $\approx$ low molecular weight
 - b) Highly crystalline $\approx$ low crystallinity, high molecular weight $>$ low molecular weight
 - c) Highly crystalline $>$ low crystallinity, high molecular weight $>$ low molecular weight
 - d) Highly crystalline $>$ low crystallinity, high molecular weight $\approx$ low molecular weight

2. You are to choose the best material for use as a load bearing hinge surface. Without knowing anything else, which of the following polymers is the most appropriate for producing these moulds? Which is the least?
 - a) Low molecular weight, highly branched polyethylene <- **LEAST**
 - b) High molecular weight, and lightly branched polyethylene
 - c) Ultrahigh molecular weight and highly branched polyethylene
 - d) Ultrahigh molecular weight and lightly branched polyethylene <- **MOST**

3. A hypothetical polymer has the following molecular weight distribution. Estimate the weight average molecular weight of this polymer

Mean Molecular Weight (g/mol)	Number Fraction
7300	0.05
23200	0.17
28000	0.48

 - a) 36246.3 g/mol
 - b) 11477.9 g/mol
 - c) 32209.0 g/mol
 - d) 12587.0 g/mol

Practice Quiz

4. Which general class of material has a Young's modulus that is roughly one hundred times less than that of metals?
 - a) Polymers
 - b) Ceramics
 - c) Composites
 - d) None of the above
5. For each of the following polymer pairs, state which you would expect to crystallize to a higher extent compared to the other. Justify your choice.
 - a) Syndiotactic linear polyethylene vs. isotactic polystyrene
 - b) Linear polyethylene vs. atactic polypropylene

NOTE: Found out this is in the videos but not the text, will cover this in more detail next tutorial but view the link below if you want an overview

<https://www.differencebetween.com/difference-between-atactic-isotactic-and-syndiotactic-polymer/>